

Classic Models Sales Dashboard

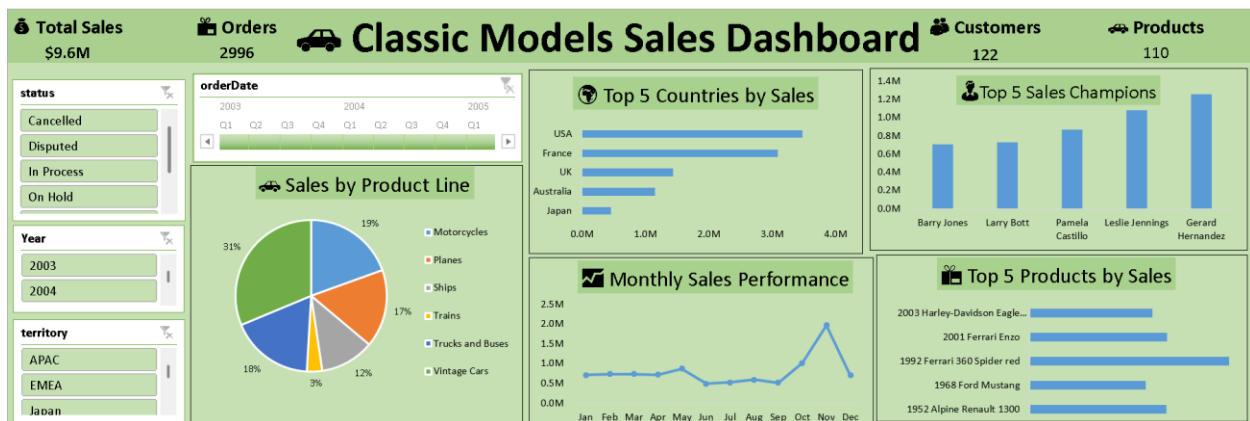
Project Report — Excel Data Model, Power Pivot & Interactive Dashboard

Overview

This project turns raw sales data — customers, orders, order details, employees, offices, products, and payments — into a single interactive Excel dashboard. It uses Power Pivot and the Excel Data Model to connect all the underlying tables, so a handful of pivot tables and pivot charts can summarize the entire business at a glance.

The dashboard lets a viewer filter by order status, region, and year, and instantly see how sales break down by country, employee, product line, and month.

Dashboard at a Glance



Key Metrics

\$9,604,190.61 Total Sales	2,996 Orders	122 Customers	110 Products
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What the Dashboard Shows

- Top 5 Countries by Sales — USA and France lead the way, followed by the UK, Australia, and Japan.
- Top 5 Sales Champions — a ranked view of the highest-performing employees by total sales, with Gerard Hernandez currently on top.
- Sales by Product Line — a pie chart breaking down revenue across all six product lines: Vintage Cars, Motorcycles, Trucks and Buses, Planes, Ships, and Trains.
- Monthly Sales Performance — a trend line across the year showing a clear seasonal spike in November.
- Top 5 Products by Sales — the best-selling individual products, led by the 1992 Ferrari 360 Spider Red.
- Interactive filters — slicers for order Status, Territory, and Year let anyone reshape every chart at once, without touching a formula.

Key Insights

- The USA and France together account for a large share of total sales, with the next three countries (UK, Australia, Japan) trailing well behind.
- There is a strong seasonal spike in November — sales roughly double the typical monthly level, likely tied to holiday-season ordering.
- Sales performance is fairly concentrated at the top: a small group of employees (led by Gerard Hernandez) drives a disproportionate share of revenue.
- Vintage Cars is the strongest product line by revenue share, at 31% of total sales — with Motorcycles (19%) and Trucks and Buses (18%) as the next largest lines. The single best-selling product overall is the 1992 Ferrari 360 Spider Red.

Behind the Scenes: Building the Data Model

Before the dashboard could work, all the source tables had to be connected correctly through Power Pivot relationships, for example:

- Customers → Orders → Order Details, so that a sale can be traced back to the customer who placed it.
- Employees → Customers, so that each customer's sales roll up to the employee who manages that account.
- Products → Product Lines, and Offices → Employees, to support the product and regional views.

Along the way, a data-quality issue was tracked down and fixed: one relationship was accidentally linking an order's customer number to a customer's name field instead of their customer number. Because a number can never match a name, that single broken link was silently blocking sales data from ever reaching the Country and Employee reports — both of which showed blank results until it was corrected.

Once the relationship was fixed to properly match customer numbers on both sides, every dependent report — Country sales, Employee performance, and everything downstream — began populating correctly.

Relationship Fix — Before and After

Before the fix: the relationship list shows an active but incorrect link between an order's customer number and the customer's name field.

Active	Table	Related Lookup Table
Yes	Table1 [officeCode]	Table3 [officeCode]
Yes	Table2 [salesRepEmployeeNumber]	Table1 [employeeNumber]
Yes	Table4 [orderNumber]	Table5 [orderNumber]
Yes	Table4 [productCode]	Table8 [productCode]
No	Table4 [sales]	Table1 [FullName2]
Yes	Table5 [customerNumber]	Table2 [customerName]
Yes	Table6 [customerNumber]	Table2 [customerNumber]
Yes	Table8 [productLine]	Table7 [productLine]

After the fix: the relationship correctly matches customer number to customer number, allowing sales to flow through to every dependent report.

Active	Table	Related Lookup Table
Yes	Table1 [officeCode]	Table3 [officeCode]
Yes	Table2 [salesRepEmployeeNumber]	Table1 [employeeNumber]
Yes	Table4 [orderNumber]	Table5 [orderNumber]
Yes	Table4 [productCode]	Table8 [productCode]
Yes	Table5 [customerNumber]	Table2 [customerName]
Yes	Table6 [customerNumber]	Table2 [customerNumber]
Yes	Table8 [productLine]	Table7 [productLine]

Result: Employee Performance Report

With the relationship corrected, the Employee Performance report immediately began showing real names and accurate sales totals, matching the overall grand total exactly.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1													
2													
3	Row Labels	Sum of sales											
4	Andy Fixter	562582.59											
5	Barry Jones	704853.91											
6	Foon Yue Tseng	488212.67											
7	George Vanauf	669377.05											
8	Gerard Hernandez	1258577.81											
9	Julie Firrelli	386663.2											
10	Larry Bott	732096.79											
11	Leslie Jennings	1081530.54											
12	Leslie Thompson	347533.03											
13	Loui Bondur	569485.75											
14	Mami Nishi	457110.07											
15	Martin Gerard	387477.47											
16	Pamela Castillo	868220.55											
17	Peter Marsh	584593.76											
18	Steve Patterson	505875.42											
19	Grand Total	9604190.61											
20													
21													

PivotTable Fields

ACTIVE ALL

Choose fields to add to report:

Table1

- employeeNumber
- lastName
- firstName
- full name
- extension

Drag fields between areas below:

FILTERS

COLUMNS

ROWS

VALUES

FullName2

Sum of sales

Skills Demonstrated

- Excel Data Model & Power Pivot — building relationships across multiple related tables.
- Data troubleshooting — systematically isolating the root cause of a data issue across several possible explanations.
- PivotTables & PivotCharts — summarizing large datasets into clear, rankable views.
- Dashboard design — combining KPIs, charts, and interactive slicers into a single cohesive, decision-ready view.

Closing Note

This dashboard is built on the widely-used "Classic Models" sample sales dataset, commonly used for practicing data analysis and business intelligence skills. It reflects a complete, working example of connecting raw relational data to a polished, interactive business dashboard in Excel.